

EQUIPMENT LIST

1

COMPRESSOR

2

FLOODED EVAPORATOR (SHELL & PLATE)

3

MULTI CONDENSER

4

OIL SEPARATOR

5

OIL COOLER

6

OIL FILTER

7

WATER INLET TEMPERATURE SENSOR

8

WATER OUTLET TEMPERATURE SENSOR

9

CRANKCASE HEATER (BUILT-IN)

10

MANUAL OIL DRAIN VALVE

11

SERVICE VALVE/ PURGE VALVE

12

SAFETY VALVE HIGH PRESSURE SIDE

13

SAFETY VALVE LOW PRESSURE SIDE

14

ELECTRONIC EXPANSION VALVE

15

FLOW SWITCH WATER SIDE

16

PRESSURE TRANSDUCER WATER SIDE

17

STRAINER

18

CHECK VALVE

19

DRAIN VALVE FOR WATER SIDE

20

LIQUID LEVEL SENSOR

21

OIL HEATER WITH THERMOSTAT (20A) AND TEMPERATURE LIMITER (20B)

22

SOLENOID VALVE

23

RATED CURRENT LIMITER & THERMAL WINDING PROTECTION-COMPRESSOR DRIVING MOTOR

24

VAPOR QUALITY SENSOR (HB)

25

OIL LEVEL SENSOR (HB)

26

TWO WAY CONTROL VALVE (MODULATING)

27

PRESSURE SENSOR

28

TEMPERATURE SENSOR

29

DISCHARGE ANGLE STOP VALVE

30

SUCTION ANGLE STOP VALVE

31

OIL SIDE ANGLE STOP VALVE

32

LIQUID LINE ANGLE STOP VALVE

33

DUAL PRESSURE SWITCH

34

CHECK VALVE

35

LIQUID LEVEL PROBE(HB)

36

STRAIGHT STOP VALVE

37

OIL EJECTOR

38

OIL CONTROL VALVE

39

FLOAT EXPANSION VALVE

L1

DISCHARGE LINE

L2

SUCTION LINE

L3

LIQUID LINE

L5

OIL RETURN LINE

SYMBOL & LEGENDS

COMPRESSOR

PURGE VALVE/SERVICE VALVE

PIPE INSULATION

TEMPERATURE TRANSDUCER

PRESSURE TRANSDUCER

BALL VALVE (STRAIGHT)

BALL VALVE (ANGLE)

SOLENOID VALVE

PRESSURE SWITCH

PRV (PRESSURE RELIEF VALVE)

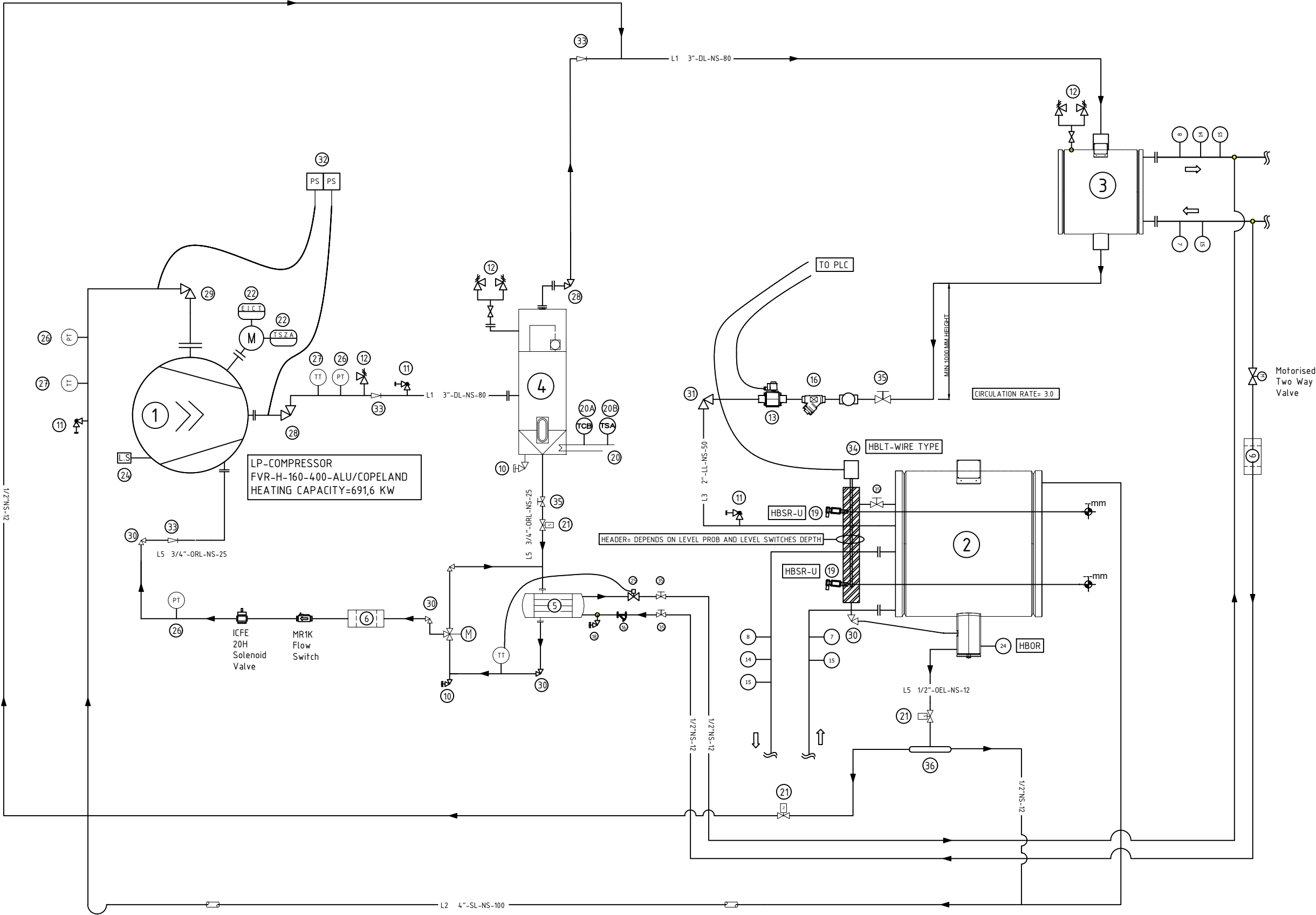
CHECK VALVE/NON RETURN VALVE

SIGHT GLASS

AIR VENT (WATER SIDE)

3 WAY VALVE

FLOW SWITCH



PIPING AND INSTRUMENTATION DIAGRAM

PROJECT NAME:	690KW HEAT PUMP (R717)	ffiTec HEATING + COOLING
PROJECT NUM:	2604-1	
REV:	1	WATER TO WATER HEAT PUMP(R717)
DATE:	20-APR-26	REFRIGERATION PIPING AND INSTRUMENTATION DIAGRAM